agent: discovery-backend-modernization-agent cli: Claude Code CLI llm: claude-sonnet-4-6 run_id: 20260826T162205_bs3b51 generated_at: 2026-08-26T10:52:11.702Z

4. Backend Discovery & Modernization Analysis

Objective: Comprehensive backend discovery covering architecture, modules, controller/service/repository layering, database schema, API governance, middleware, authentication & authorization, security, performance, dependencies, secrets, and code quality.

Date: 2026-08-26 16:24:18 IST | **Scope:** `shende-shweta/pingcrm` — PHP 8.2 / Laravel 11.x with Inertia.js + React/TypeScript frontend

Executive Summary

EXECUTIVE SUMMARY

The PingCRM repository is a bifurcated codebase: a clean Laravel 11 CRM core (Contacts, Organizations, Users, Reports) sharing its scaffolding with a deeply problematic Legacy IVR Enterprise module that was bolted on in mid-2026. The Legacy IVR layer contains 12 "GodService" classes (one per module) each carrying 45 near-identical workflow methods that use PHP's `extract()` to materialize raw payload fields as local variables, hold a mutable static cache array, make unparameterized string-concatenated SQL via the `DB` facade, and `sleep(1)` on every call — representing over 540 unsafe `extract()` occurrences and 540 synchronous blocking seconds per full workflow cycle. Twelve parallel repository files compound this with explicit SQL-injection-vulnerable string-concatenated `LIKE` queries. The `config/ivr_legacy.php` config file commits a Salesforce client secret, plain-text password, and a master API key directly to source control, while all 12 GodService files each hardcode their own `LEGACY_IVR_KEY`. The generated legacy API surface accepts both `GET` and `POST` on all state-mutating endpoints without any authentication middleware, creating an unauthenticated write path to IVR data for any network-reachable client. Overall backend health is **High Risk**, driven primarily by widespread SQL injection, hundreds of hardcoded secrets, mass-assignment-open Eloquent models, missing authentication on the generated API surface, and synchronous blocking I/O in every legacy workflow.

91	50+	12	~110	15+	0
CONTROLLERS / HANDLERS SCANNED	FILES USING DYNAMIC-VARIABLE PATTERNS	LEGACY GODSERVICE CLASSES FOUND	API ENDPOINTS FOUND	SECURITY RISK PATTERNS FOUND	CRITICAL / HIGH CVES FOUND (AUDIT NOT RUN)

OVERALL CODEBASE RATING — BACKEND MODERNIZATION

High Risk

> Driven by H1 (4,940+ extract() occurrences), H13 (SQL injection in 12 repositories + mass assignment), H16 (secrets committed in source), H12 (unauthenticated generated API surface), and H14 (540 sleep(1) blocking calls).

4.1 Benchmark Ratings Summary

One row per hotspot. "Measured" is the real value found; "Rating" is the band it falls into (worst KPI wins).

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	Dynamic Variable Creation	Dynamic-var-from-input occurrences	0	1–10	>10	4,940+ `extract()` calls across 50+ files	HIGH RIS
H2	Global Mutable State	Globals / mutable static state	0	1–5	>5	12 `public static $sharedRuntimeCache` (one per GodService)	HIGH RIS
H3	Direct SQL Outside Data Layer	Data-layer compliance %	>90%	60–90%	<60%	~30% — 1,560 raw DB calls; controllers, traits, and services bypass repository	HIGH RIS
H4	Static / Singleton Abuse	Business-logic static/singleton classes	0	1–5	>5	5 pure-static helper classes (LegacyIvrCrypto with 80 methods, LegacyIvrArray, LegacyIvrDate, LegacyIvrMath, LegacyIvrString)	HIGH RIS
H5	Missing Service Layer	Handlers with inline business logic	<10	10–20	>20	70+ Ivr controllers contain inline SQL queries, direct service instantiation, and business logic	HIGH RIS
H6	API Sprawl	Documented & governed endpoints %	>90%	80–90%	<80%	<10% — ~90 generated routes accept GET+POST; no versioning, no spec	HIGH RIS
H7	Missing API Governance	Governance compliance %	100%	90–99%	<90%	0% — No OpenAPI spec, no API versioning, no contract tests	HIGH RIS
H8	Weak Application Architecture	Modules following declared architecture %	>80%	50–80%	<50%	~40% — CRM core follows MVC cleanly; all 70+ IVR controllers are fat handlers with inline queries	HIGH RIS
H9	Missing Module Inventory	Circular dependency count	0	1–3	>3	0 circular deps detected; Legacy and Http\Controllers\Ivr are	MODERA

						tightly coupled with no documented module API	
H10	Database Schema Weakness	FK indexes % + migrations with rollback %	Both >90%	One <90%	Both <90%	FK indexes ~60% (IVR tables lack FK constraints); rollback: 100% (all migrations have down())	MODERA
H11	Middleware Weakness	Required middleware present + ordered %	100%	80–99%	<80%	~55% — /img/{path} unprotected; entire generated ivr-legacy API surface has no auth middleware	HIGH RIS
H12	Auth & Authorization Weakness	Protected routes guarded % + hashing algo	100% + bcrypt/argon2	One gap	Both bad	~60% routes guarded; password hashing: bcrypt (good); $proxies='*' enables IP spoofing; tenant_id=1 hardcoded	HIGH RIS
H13	Backend Security Vulnerabilities	Injection + hardcoded secrets count	0 each	1–3 total	>3 total	SQL injection: 960 patterns; mass assignment: 12 models $guarded=[]; hardcoded secrets: 15+	HIGH RIS
H14	Performance & Caching Gaps	N+1 patterns found	0	1–5	>5	540 sleep(1) blocking calls; 35+ N+1-prone model accessors; no caching layer	HIGH RIS
H15	Outdated & Vulnerable Dependencies	Critical/High CVEs found	0	1–3	>3	roave/security-advisories present (CVE guard); no composer audit output; PHPStan at level 1	MODERA
H16	Secrets & Configuration in Source	Hardcoded secrets / .env committed	0	1–2	>2	15+ secrets: 12 LEGACY_IVR_KEY values + config/ivr_legacy.php master key, SF credentials, plain-text password	HIGH RIS
H17	Backend Code Quality	Linter in CI + max cyclomatic complexity	Both good	One gap	Both bad	CI exists but PHPStan at level 1; GodService has 45 identical methods per class; extreme duplication	MODERA

No additional hotspots beyond the standard set were observed.

4.2 Hotspot-by-Hotspot Evidence

H1. Dynamic Variable Creation CRITICAL

Benchmark: Dynamic-var-from-input occurrences = **4,940+** → falls in the **High Risk** band (Good 0 · Moderate 1–10 · High Risk >10)

All 12 GodService classes use PHP's `extract($payload)` on every workflow method, materializing raw array keys — including any key an attacker submits — as local PHP variables before the extracted `$tenant_id` is used without validation to key the static cache.

```
// app/Legacy/Services/AgentDeskGodService.php:13-18
public function orchestrateAgentDeskWorkflow1($payload)
{
    extract($payload); // unsafe — any key in $payload becomes a local variable
    sleep(1);
    self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload;
    return DB::table("ivr_agent_desks")->insertGetId((array) $payload);
}
```

The identical `extract($payload)` pattern repeats across all 45 methods of all 12 services (540 occurrences in Legacy/Services). Controllers also call `extract($payload)` in their `legacyEndpointN()` methods — 4,400 additional occurrences in Http/Controllers/Ivr:

```
// app/Http/Controllers/Ivr/AgentDeskStoreController.php (legacyEndpoint1
pattern)
$payload = $request->all();
extract($payload); // 4,400+ occurrences in Ivr controllers
$service = new AgentDeskGodService();
$service->orchestrateAgentDeskWorkflow1($payload);
```

Why it matters here: An attacker posting a key named `service` or `q` can silently overwrite the same-named local variable, leading to logic bypass. Combined with `(array) $payload` passed directly to `insertGetId`, any submitted field lands verbatim in the database row with no field whitelist.

Recommended approach:

1. Replace `extract($payload)` with explicit typed DTOs or `$payload['field_name']` access in all GodService methods.
2. Introduce a `StoreAgentDeskRequest` (Laravel Form Request) to whitelist allowed fields before passing to service.
3. Add PHPStan custom rule `no-extract` to prevent regression.
4. Remove `legacyEndpointN()` controller methods — they duplicate GodService logic without validation.

92 files affected.

File	Line(s)	Issue	Action
`app/Http/Controllers/Ivr/AgentDeskDestroyController.php`	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

app/Http/Controllers/Ivr/AgentDeskExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/AgentDeskImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/AgentDeskIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/AgentDeskStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/AgentDeskSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/AgentDeskUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/BusinessHoursDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/BusinessHoursExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/BusinessHoursImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

	178, 191 +43		
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallFlowDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallFlowExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallFlowImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallFlowIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallFlowStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallFlowSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

	178, 191 +43		
app/Http/Controllers/Ivr/CallFlowUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRecordingExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRecordingImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRecordingIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRecordingStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRecordingSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

app/Http/Controllers/Ivr/CallRoutingDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRoutingExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRoutingImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRoutingIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRoutingStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRoutingSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

	178, 191 +43		
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/DidInventoryExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/DidInventoryImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/DidInventoryIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/DidInventoryStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/DidInventorySyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

	178, 191 +43		
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/PromptLibraryExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/PromptLibraryImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

app/Http/Controllers/Ivr/PromptLibraryIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/QueueManagementExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/QueueManagementImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/QueueManagementIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/QueueManagementStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165,	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

	178, 191 +43		
app/Http/Controllers/Ivr/QueueManagementSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Http/Controllers/Ivr/QueueManagementUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Legacy/Services/AgentDeskGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Legacy/Services/BusinessHoursGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Legacy/Services/CallAnalyticsGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Legacy/Services/CallFlowGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Legacy/Services/CallRecordingGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
app/Legacy/Services/CallRoutingGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

`app/Legacy/Services/CustomerProfileGodService.php`	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
`app/Legacy/Services/DidInventoryGodService.php`	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
`app/Legacy/Services/HistoricalReportsGodService.php`	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
`app/Legacy/Services/LiveMonitoringGodService.php`	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
`app/Legacy/Services/PromptLibraryGodService.php`	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation
`app/Legacy/Services/QueueManagementGodService.php`	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Uses extract() to materialize raw input as local variables — variable shadowing and injection risk	Replace with explicit DTO or field-by-field access; add Form Request validation

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H2. Global Mutable State CRITICAL

Benchmark: Mutable static state holding business data = **12 instances** → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5)

Every GodService class declares a public static array used as a per-request cache keyed by tenant ID. In PHP-FPM, workers are reused across requests, so this cache can leak data from one request to another.

```
// app/Legacy/Services/AgentDeskGodService.php:9-11
class AgentDeskGodService
{
    public static $sharedRuntimeCache = []; // mutable global-ish state
    private $apiKey = "LEGACY_IVR_KEY_2042"; // hard-coded secret
```

Same pattern in all 12 services: `BusinessHoursGodService`, `CallAnalyticsGodService`, `CallFlowGodService`, `CallRecordingGodService`, `CallRoutingGodService`, `CustomerProfileGodService`, `DidInventoryGodService`, `HistoricalReportsGodService`, `LiveMonitoringGodService`, `PromptLibraryGodService`, `QueueManagementGodService`.

Why it matters here: In a multi-tenant IVR system, tenant A's payload cached under tenant ID 1 during one request can be read by tenant B's request served by the same worker before the cache is overwritten — a direct data-isolation violation. The `public` visibility also means any code anywhere can read or modify the cache.

Recommended approach:

1. Remove `public static $sharedRuntimeCache` from all 12 GodService classes.
2. If per-request memoization is needed, use an instance property or Laravel's `Request`-scoped container binding.
3. Add a PHPStan rule banning public static mutable arrays on service classes.

12 files affected.

File	Line(s)	Issue	Action
`app/Legacy/Services/AgentDeskGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/BusinessHoursGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/CallAnalyticsGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/CallFlowGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/CallRecordingGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/CallRoutingGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/CustomerProfileGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/DidInventoryGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/HistoricalReportsGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/LiveMonitoringGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/PromptLibraryGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding
`app/Legacy/Services/QueueManagementGodService.php`	10	Public static mutable cache enables cross-request data leakage in PHP-FPM	Convert to per-request scoped instance property or Laravel container binding

H3. Direct SQL / ORM Outside Data Layer CRITICAL

Benchmark: Data-layer compliance % = **~30%** (1,560 raw DB calls; controllers, traits, and service methods bypass the repository layer) → falls in the **High Risk** band (Good >90% · Moderate 60–90% · High Risk <60%)

`IvrHubController`, `ReportsController`, `LoadsIvrModuleData` trait, and all 12 GodService classes issue `DB::table()`, `DB::select()`, and `DB::raw()` calls directly, bypassing the `App\Repositories\Legacy\*` classes:

```
// app/Http/Controllers/Ivr/IvrHubController.php:55-57
private function loadStats(IvrAccountContext $ctx, array $filters): array
{
    $queueQuery = DB::table('ivr_operational_queues')->where('account_id',
$ctx->accountId);
    $agentsQuery = DB::table('ivr_agents')->where('account_id', $ctx-
>accountId);
```

```
// app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php:147-152
protected function loadConfigRows(string $module, IvrAccountContext $ctx,
string $q): array
{
    $table = $this->tableForModule($module);
    $query = DB::table($table)->where('account_id', $ctx->accountId);
```

GodService classes also call `DB::table()` in every method instead of delegating to the repository:

```
// app/Legacy/Services/AgentDeskGodService.php:18
return DB::table("ivr_agent_desks")->insertGetId((array) $payload);
```

Why it matters here: The repository layer exists (`app/Repositories/Legacy/`) but is bypassed 70%+ of the time. Any schema change (e.g., adding a soft-delete filter) must be applied across all DB call sites — a maintenance and correctness risk.

Recommended approach:

1. Move all `DB::table()` calls in `IvrHubController` and `ReportsController` into dedicated `IvrCallRepository`, `IvrQueueRepository` classes.
2. Enforce via PHPStan custom rule: `DB::table()` banned in `Http\Controllers\*` namespace.
3. Refactor GodService methods to delegate all persistence to the corresponding `App\Repositories\Legacy\*` repository.
4. Add eager-loading specifications to repositories to prevent N+1.

120 files affected.

File	Line(s)	Issue	Action
`app/Http/Controllers/Ivr/AgentDeskDestroyController.php`	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

app/Http/Controllers/Ivr/AgentDeskExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/AgentDeskImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/AgentDeskIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/AgentDeskStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/AgentDeskSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/AgentDeskUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/BusinessHoursDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/BusinessHoursExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/BusinessHoursImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallFlowDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallFlowExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallFlowImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallFlowIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallFlowStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallFlowSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallFlowUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRecordingExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRecordingImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRecordingIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRecordingStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRecordingSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRoutingExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRoutingImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRoutingIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

app/Http/Controllers/Ivr/CallRoutingStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRoutingSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php	82, 110, 140, 168, 185, 204	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/DidInventoryExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/DidInventoryImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/DidInventoryIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/DidInventoryStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/DidInventorySyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/IvrHubController.php	68, 77, 92, 132, 148, 167, 204, 221, 227, 232, 240, 258 +4	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/PromptLibraryExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/PromptLibraryImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized

			bindings
app/Http/Controllers/Ivr/QueueManagementExportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/QueueManagementImportController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/QueueManagementIndexController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/QueueManagementStoreController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/QueueManagementSyncController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/Ivr/QueueManagementUpdateController.php	28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Http/Controllers/ReportsController.php	61, 77, 98, 115, 166	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/AgentDeskGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/BusinessHoursGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/CallAnalyticsGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/CallFlowGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/CallRecordingGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

app/Legacy/Services/CallRoutingGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/CustomerProfileGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/DidInventoryGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/HistoricalReportsGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/LiveMonitoringGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/PromptLibraryGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Legacy/Services/QueueManagementGodService.php	18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106 +33	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/AgentDesk.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/BusinessHours.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79,	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

	85, 91 +23		
app/Models/Ivr/CallAnalytics.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/CallFlow.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/CallRecording.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/CallRouting.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/CustomerProfile.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/DidInventory.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/HistoricalReports.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/LiveMonitoring.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Models/Ivr/PromptLibrary.php	25, 31, 37, 43, 49, 55, 61, 67,	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

	73, 79, 85, 91 +23		
app/Models/Ivr/QueueManagement.php	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Repositories/Legacy/AgentDeskRepository.php	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Repositories/Legacy/BusinessHoursRepository.php	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Repositories/Legacy/CallAnalyticsRepository.php	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Repositories/Legacy/CallFlowRepository.php	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Repositories/Legacy/CallRecordingRepository.php	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Repositories/Legacy/CallRoutingRepository.php	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
app/Repositories/Legacy/CustomerProfileRepository.php	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

`app/Repositories/Legacy/DidInventoryRepository.php`	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
`app/Repositories/Legacy/HistoricalReportsRepository.php`	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
`app/Repositories/Legacy/LiveMonitoringRepository.php`	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
`app/Repositories/Legacy/PromptLibraryRepository.php`	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
`app/Repositories/Legacy/QueueManagementRepository.php`	16, 25, 34, 43, 52, 61, 70, 79, 88, 97, 106, 115 +28	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings
`app/Support/IvrAccountContext.php`	62	Raw DB query outside repository/data-access layer	Move query to Repository class; use parameterized bindings

---

H4. Static Methods & Singleton Abuse HIGH

Benchmark: Business-logic static/singleton classes = **5 pure-static helper classes** → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5)

The entire `App\Legacy\Helpers` namespace consists of five classes (`LegacyIvrCrypto` , `LegacyIvrArray` , `LegacyIvrDate` , `LegacyIvrMath` , `LegacyIvrString`) that are 100% static methods. `LegacyIvrCrypto` alone has 80 static `transform` methods, all performing identical string concatenation:

```
// app/Legacy/Helpers/LegacyIvrCrypto.php:4-12
class LegacyIvrCrypto
{
    public static function transform1($value)
    {
        // duplicate of other helper — kept for backward compatibility
        if ($value === null) { return ""; }
```

```
        return (string) $value . "_2130_1";
    }
    // ... transform2() through transform80() — identical logic, different
suffix
}
```

Why it matters here: Static utility classes cannot be injected as dependencies, cannot be mocked in unit tests, and accumulate duplicate methods. The 80 `transform*` methods in `LegacyIvrCrypto` are doing the same string suffix operation — an entire class that could be a single parameterized function.

Recommended approach:

1. Consolidate `LegacyIvrCrypto::transform1()...transform80()` into a single `transform(string $value, int $index): string` method.
2. Register helpers as Laravel service container singletons if stateless utility is required, enabling mock injection in tests.
3. Convert multi-method duplicate helper classes into injectable services with a single implementation.

5 files affected.

File	Line(s)	Issue	Action
`app/Legacy/Helpers/LegacyIvrArray.php`	7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84 +68	Pure-static helper class with 80 near-identical methods prevents DI and mocking	Consolidate to single parameterized method; register as injectable service
`app/Legacy/Helpers/LegacyIvrCrypto.php`	7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84 +68	Pure-static helper class with 80 near-identical methods prevents DI and mocking	Consolidate to single parameterized method; register as injectable service
`app/Legacy/Helpers/LegacyIvrDate.php`	7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84 +68	Pure-static helper class with 80 near-identical methods prevents DI and mocking	Consolidate to single parameterized method; register as injectable service
`app/Legacy/Helpers/LegacyIvrMath.php`	7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84 +68	Pure-static helper class with 80 near-identical methods prevents DI and mocking	Consolidate to single parameterized method; register as injectable service
`app/Legacy/Helpers/LegacyIvrString.php`	7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84 +68	Pure-static helper class with 80 near-identical methods prevents DI and mocking	Consolidate to single parameterized method; register as injectable service

---

H5. Missing Service Layer CRITICAL

Benchmark: Handlers with inline business logic = **70+ Ivr controllers** → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20)

The 70+ `Ivr` single-action controllers each contain inline business logic: raw SQL queries, direct service instantiation with `new AgentDeskGodService()`, hard-coded tenant IDs, and error swallowing with empty catch blocks:

```
// app/Http/Controllers/Ivr/AgentDeskStoreController.php:14-38
class AgentDeskStoreController extends Controller
{
    // AUTH-NOTE: some endpoints intentionally skip policies (2014 regression)
    private $tenantId = 1; // hard-coded tenant — multi-tenant broken

    public function handleStore(Request $request)
    {
        $service = new AgentDeskGodService(); // manual instantiation, no DI
        $q = $request->get("q");
        if ($q) {
            $rows = DB::select("select * from ivr_agent_desks where name like
'%".$q."%' and tenant_id = ".$this->tenantId);
        } else {
            $rows = AgentDesk::where("tenant_id", $this->tenantId)->get();
        }
        // ...
    }

    public function legacyEndpoint1(Request $request)
    {
        try {
            $payload = $request->all();
            extract($payload);
            $service = new AgentDeskGodService();
            $service->orchestrateAgentDeskWorkflow1($payload);
            return ["ok" => true, "endpoint" => 1];
        } catch (\Throwable $e) {
            return ["ok" => false, "err" => $e->getMessage()]; // swallowed
stack traces
        }
    }
}
```

Why it matters here: With business logic in 70+ controllers, the same validation/filtering/persistence logic cannot be reused from CLI commands, queue jobs, or API endpoints without duplicating the controller. A single multi-tenant correction (e.g., correctly scoping by `account_id`) must be applied in 70+ places independently.

Recommended approach:

1. Extract all `AgentDesk*Controller` logic into an `AgentDeskService` that accepts typed DTOs.
2. Repeat for all 11 other IVR modules (one `XxxService` per module).
3. Controllers become thin: validate request → call service → return Inertia/JSON response.

4. Consolidate the 12 "GodService" classes into the proper `XxxService` classes with injected repositories.

35 files affected.

File	Line(s)	Issue	Action
`app/Http/Controllers/Ivr/AgentDeskDestroyController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
`app/Http/Controllers/Ivr/AgentDeskExportController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
`app/Http/Controllers/Ivr/AgentDeskImportController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
`app/Http/Controllers/Ivr/AgentDeskIndexController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
`app/Http/Controllers/Ivr/AgentDeskStoreController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
`app/Http/Controllers/Ivr/AgentDeskSyncController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
`app/Http/Controllers/Ivr/AgentDeskUpdateController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
`app/Http/Controllers/Ivr/BusinessHoursDestroyController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class

app/Http/Controllers/Ivr/BusinessHoursExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/BusinessHoursImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallFlowDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallFlowExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallFlowImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153,	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class

	166, 179 +44		
app/Http/Controllers/Ivr/CallFlowIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallFlowStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallFlowSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallFlowUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallRoutingExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallRoutingImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallRoutingIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class

app/Http/Controllers/Ivr/CallRoutingStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallRoutingSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/QueueManagementExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/QueueManagementImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/QueueManagementIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/QueueManagementStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class
app/Http/Controllers/Ivr/QueueManagementSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153,	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class

	166, 179 +44		
`app/Http/Controllers/Ivr/QueueManagementUpdateController.php`	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Direct GodService instantiation in controller — business logic not in service layer	Inject typed service via constructor DI; move business logic to service class

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H6. API Sprawl HIGH

Benchmark: Documented & governed endpoints % = **<10%** → falls in the **High Risk** band (Good >90% · Moderate 80–90% · High Risk <80%)

The generated `routes/generated/ivr_legacy_api.php` exposes every IVR module action via `Route::match(['get','post'], ...)` — meaning state-mutating operations (destroy, store, update) are accessible via GET requests. No resource conventions, versioning prefix, or documentation exist.

```
// routes/generated/ivr_legacy_api.php:5-10
Route::prefix("ivr-legacy")->group(function () {
    Route::match(['get','post'], 'agent-desk/destroy',
App\Http\Controllers\Ivr\AgentDeskDestroyController::class);
    Route::match(['get','post'], 'agent-desk/store',
App\Http\Controllers\Ivr\AgentDeskStoreController::class);
    // ... 90+ more routes with same pattern
});
```

```
// routes/api.php:7-9
Route::get('/ivr/health-legacy', function () {
    return response()->json(['status' => 'maybe-ok', 'timestamp' => time()]);
});
```

Why it matters here: Accepting GET for destroy/store operations allows CSRF via hyperlink or image tag. The `'status' => 'maybe-ok'` health check signals the team themselves are uncertain about system health.

Recommended approach:

1. Replace `Route::match(['get','post'], ...)` with appropriate HTTP verbs: `GET` for index/show, `POST` for store, `PUT/PATCH` for update, `DELETE` for destroy.
2. Add `/api/v1/` prefix to all API routes.
3. Group all IVR API routes under a version-scoped route group with auth middleware.
4. Generate an OpenAPI spec from route metadata using `dedoc/scramble`.

1 file affected.

File	Line(s)	Issue	Action
`routes/generated/ivr_legacy_api.php`	7, 8, 9, 10, 11, 12, 13,	State-mutating endpoints accept GET via Route::match — CSRF risk; no versioning	Split to proper HTTP verbs; add /api/v1/ prefix; add auth middleware

	14, 15, 16, 17, 18 +68		

H7. Missing API Governance HIGH

Benchmark: Governance compliance % = **0%** → falls in the **High Risk** band (Good 100% · Moderate 90–99% · High Risk <90%)

No OpenAPI/Swagger spec file was found. There is no API linting, no contract tests, and no changelog. The `routes/generated/ivr_legacy_api.php` is auto-generated from controller names without a schema definition.

```
// routes/api.php:3
require __DIR__.'/generated/ivr_legacy_api.php'; // generated, not schema-driven
```

Why it matters here: Without a spec, any refactor or route rename is an undetected breaking change for API consumers. Frontend Inertia components hardcode endpoint paths with no contract to validate against.

Recommended approach:

1. Add `dedoc/scramble` or `knuckleswtf/scribe` to auto-generate OpenAPI from Laravel routes.
2. Add Spectral API linting to CI (`coding-standards.yml`).
3. Add consumer-driven contract tests for the API surface.
4. Add `/api/v1/` versioning prefix before the first external consumer onboards.

3 files affected.

File	Line(s)	Issue	Action
`routes/api.php`	7	Routes registered without OpenAPI spec, versioning, or auth middleware	Generate OpenAPI spec; add versioning prefix; add auth guard to all state-mutating routes
`routes/generated/ivr_legacy_api.php`	6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17 +69	Routes registered without OpenAPI spec, versioning, or auth middleware	Generate OpenAPI spec; add versioning prefix; add auth guard to all state-mutating routes
`routes/web.php`	26, 30, 34, 39, 44, 45, 46, 47, 50, 57, 61, 65 +21	Routes registered without OpenAPI spec, versioning, or auth middleware	Generate OpenAPI spec; add versioning prefix; add auth guard to all state-mutating routes

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H8. Weak Application Architecture HIGH

Benchmark: Modules following declared architecture % = **~40%** → falls in the **High Risk** band (Good >80% · Moderate 50–80% · High Risk <50%)

The CRM core follows Laravel MVC: thin controllers with explicit validation delegate to Eloquent models. The Legacy IVR layer violates this at every level — fat controllers, direct DB calls, business logic in services that also issue raw SQL, and a

`LoadsIvrModuleData` trait carrying hundreds of lines of query logic:

```
// app/Http/Controllers/Ivr/AgentDeskStoreController.php:14
// AUTH-NOTE: some endpoints intentionally skip policies (2014 regression)
private $tenantId = 1; // hard-coded tenant — multi-tenant broken
```

The `LoadsIvrModuleData` trait carries all data-loading logic for 18 module types directly in the controller concern layer — mixing controller responsibility with data-access responsibility.

Why it matters here: New developers cannot follow consistent patterns. The ~60% of code that ignores the declared MVC architecture produces bugs at unpredictable rates — missing `account_id` scoping, bypassed auth policies, and untestable logic.

Recommended approach:

1. Define the target architecture: thin controller → Form Request → Service → Repository → Eloquent Model. Document in `ARCHITECTURE.md`.
2. Extract all IVR query logic from `LoadsIvrModuleData` into repositories.
3. Bind services via Laravel's service container and inject via constructor.
4. Implement proper Laravel policies for IVR module actions to replace the `AUTH-NOTE` regression.

80 files affected.

File	Line(s)	Issue	Action
`app/Http/Controllers/Ivr/AgentDeskDestroyController.php`	14, 15, 25, 49, 62, 75, 88, 101, 114, 127, 140, 153 +46	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskExportController.php`	14, 15, 25, 49, 62, 75, 88, 101, 114, 127, 140, 153 +46	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskImportController.php`	14, 15, 25, 49, 62, 75, 88, 101, 114, 127, 140, 153 +46	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskIndexController.php`	14, 15, 25, 49, 62, 75, 88, 101, 114, 127, 140, 153 +46	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskStoreController.php`	14, 15, 25, 49, 62, 75, 88, 101, 114, 127,	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy

	140, 153 +46		
app/Http/Controllers/Ivr/AgentDeskSyncController.php	14, 15, 25, 49, 62, 75, 88, 101, 114, 127, 140, 153 +46	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/AgentDeskUpdateController.php	14, 15, 25, 49, 62, 75, 88, 101, 114, 127, 140, 153 +46	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy

app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallFlowDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallFlowExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallFlowImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallFlowIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallFlowStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallFlowSyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallFlowUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRecordingExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRecordingImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRecordingIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service	Use constructor DI; derive tenant from authenticated

		instantiation, and bypassed auth policy	user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRecordingStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRecordingSyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRoutingExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRoutingImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRoutingIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRoutingStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRoutingSyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy

app/Http/Controllers/Ivr/DidInventoryExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/DidInventoryImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/DidInventoryIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/DidInventoryStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/DidInventorySyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	14, 15	Fat controller with hard-coded tenant, manual service	Use constructor DI; derive tenant from authenticated

		instantiation, and bypassed auth policy	user; implement Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryImportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
app/Http/Controllers/Ivr/QueueManagementExportController.php	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy

`app/Http/Controllers/Ivr/QueueManagementImportController.php`	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementIndexController.php`	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementStoreController.php`	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementSyncController.php`	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementUpdateController.php`	14, 15	Fat controller with hard-coded tenant, manual service instantiation, and bypassed auth policy	Use constructor DI; derive tenant from authenticated user; implement Laravel Policy

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H11. Middleware Weakness CRITICAL

Benchmark: Required middleware present and correctly ordered % = **~55%** → falls in the **High Risk** band (Good 100% · Moderate 80–99% · High Risk <80%)

Two significant gaps exist:

1. `/img/{path}` route has no auth middleware:

```
// routes/web.php:157-160
Route::get('/img/{path}', [ImagesController::class, 'show'])
    ->where('path', '.*')
    ->name('image')
    // ← no ->middleware('auth')
```

2. The entire generated `ivr-legacy` API prefix has no auth middleware, and `/api/ivr/health-legacy` is also unguarded:

```
// routes/api.php:3-8
require __DIR__.'/generated/ivr_legacy_api.php'; // no auth guard applied
Route::get('/ivr/health-legacy', function () {
    return response()->json(['status' => 'maybe-ok', 'timestamp' => time()]);
});
```

Rate limiting is applied via `throttleApi()` for API routes but web routes have no rate limit.

Why it matters here: The missing `auth` guard on the generated legacy API means any network-reachable client can call `POST /api/ivr-legacy/agent-desk/store` and write data. Combined with `extract($payload)` and `$guarded = []`, this is a complete unauthenticated arbitrary write path.

Recommended approach:

1. Add `->middleware('auth')` to the `/img/{path}` route, or use signed URLs for image serving.
2. Wrap `routes/generated/ivr_legacy_api.php` include in `Route::middleware(['auth:sanctum'])->group(...)`.
3. Restrict `/ivr/health-legacy` to internal monitoring or add auth.
4. Add `ThrottleRequests` middleware to web route groups.

3 files affected.

File	Line(s)	Issue	Action
`routes/api.php`	7	Routes registered without auth middleware — unauthenticated access to IVR write operations	Wrap in auth middleware group; use auth:sanctum for API routes
`routes/generated/ivr_legacy_api.php`	7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18 +68	Routes registered without auth middleware — unauthenticated access to IVR write operations	Wrap in auth middleware group; use auth:sanctum for API routes
`routes/web.php`	26, 30, 34, 45, 46, 47, 50, 57, 61, 65, 69, 73 +19	Routes registered without auth middleware — unauthenticated access to IVR write operations	Wrap in auth middleware group; use auth:sanctum for API routes

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H12. Auth & Authorization Weakness CRITICAL

Benchmark: Protected routes % = **~60%**; password hashing = bcrypt (good) → falls in the **High Risk** band (routing gap is severe)

Password hashing uses `Hash::make()` (bcrypt, 12 rounds) — correct. However `TrustProxies` is configured to trust `$proxies = '*'` (all proxies), and `config/ivr_legacy.php` defines an IP-based auth bypass that can be spoofed via `X-Forwarded-For`:

```
// app/Http/Middleware/TrustProxies.php:14
protected $proxies = '*'; // trusts ALL proxies — X-Forwarded-For spoofable
```

```
// config/ivr_legacy.php:14
'bypass_auth_for_internal_ips' => ['127.0.0.1', '10.0.0.0'],
```

There is also no object-level authorization for IVR records — any authenticated user can operate on any tenant's data since `$tenantId = 1` is hardcoded:

```
// app/Http/Controllers/Ivr/AgentDeskStoreController.php:15
private $tenantId = 1; // hard-coded tenant — multi-tenant broken
```

Why it matters here: `X-Forwarded-For: 127.0.0.1` + wildcard proxy trust potentially triggers the IP bypass for any internal-IP-privileged operations. Combined with the unguarded legacy API surface, attackers have two distinct unauthenticated attack

paths.

Recommended approach:

1. Replace `$proxies = '*'` with explicit load-balancer IP list from infrastructure config.
2. Remove the IP-bypass config or reimplement using signed tokens.
3. Add `Gate::authorize()` or Laravel Policy checks scoped to `account_id` in all IVR controllers.
4. Replace `private $tenantId = 1` with `Auth::user()->account_id` throughout all IVR controllers.

81 files affected.

File	Line(s)	Issue	Action
`app/Http/Controllers/Ivr/AgentDeskDestroyController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskExportController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskImportController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskIndexController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskStoreController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskSyncController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/AgentDeskUpdateController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/BusinessHoursDestroyController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/BusinessHoursExportController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/BusinessHoursImportController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/BusinessHoursIndexController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/BusinessHoursStoreController.php`	15	Wildcard proxy trust + IP bypass config enables auth	Restrict trusted proxies to known IPs; remove IP

		bypass via X-Forwarded-For spoofing	bypass; use Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallFlowDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallFlowExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallFlowImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallFlowIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallFlowStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy

app/Http/Controllers/Ivr/CallFlowSyncController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallFlowUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRecordingExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRecordingImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRecordingIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRecordingStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRecordingSyncController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRoutingExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRoutingImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRoutingIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRoutingStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CallRoutingSyncController.php	15	Wildcard proxy trust + IP bypass config enables auth	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy

		bypass via X-Forwarded-For spoofing	
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/DidInventoryExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/DidInventoryImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/DidInventoryIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/DidInventoryStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/DidInventorySyncController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy

app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryExportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryImportController.php	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	15	Wildcard proxy trust + IP bypass config enables auth	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy

		bypass via X-Forwarded-For spoofing	
`app/Http/Controllers/Ivr/PromptLibraryStoreController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/PromptLibrarySyncController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/PromptLibraryUpdateController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementDestroyController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementExportController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementImportController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementIndexController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementStoreController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementSyncController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Controllers/Ivr/QueueManagementUpdateController.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy
`app/Http/Middleware/TrustProxies.php`	15	Wildcard proxy trust + IP bypass config enables auth bypass via X-Forwarded-For spoofing	Restrict trusted proxies to known IPs; remove IP bypass; use Laravel Policy

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H13. Backend Security Vulnerabilities CRITICAL

Benchmark: Injection-risk patterns + hardcoded secrets = **975+ total** → falls in the **High Risk** band (Good 0 each · Moderate 1–3 total · High Risk >3 total)

SQL Injection (960 patterns in 12 Repository files): All 12 `App\Repositories\Legacy\*Repository` files build SQL by string concatenation with unescaped user input:

```
// app/Repositories/Legacy/AgentDeskRepository.php:10-17
public function fetchChunk1($tenantId, $filter = null)
{
    $sql = "SELECT * FROM ivr_agent_desks WHERE tenant_id = " . (int)
$tenantId;
    if ($filter) {
        $sql .= " AND name LIKE '%" . $filter . "%'"; // SQL injection —
unescaped user input
    }
    return DB::select($sql);
}
// 39 more identical fetchChunkN() methods in this file; repeated across 12
repository files
```

Mass Assignment (IVR models): All 12 IVR Eloquent models declare `protected $guarded = []`:

```
// app/Models/Ivr/AgentDesk.php:10
protected $guarded = []; // legacy — mass assignment wide open
```

Committed debug config:

```
// config/ivr_legacy.php:13
'allow_sql_debug' => true, // must never be true in production
```

Why it matters here: The SQL injection in repositories affects every `fetchChunkN()` call path. An attacker controlling the `?q=` filter parameter can exfiltrate or modify the entire `ivr_agent_desks` table (and all 11 sibling tables) using a single LIKE-injection payload like `%' OR 1=1 --`.

Recommended approach:

1. Replace all string-concatenated SQL with parameterized bindings: `DB::table('ivr_agent_desks')->where('name', 'like', '%'.$filter.'%')`.
2. Replace `$guarded = []` with explicit `$fillable` arrays on all IVR models.
3. Set `allow_sql_debug` via environment variable defaulting to `false`.
4. Add PHPStan rule and CI grep to block raw string SQL concatenation patterns.

12 files affected.

File	Line(s)	Issue	Action
`app/Repositories/Legacy/AgentDeskRepository.php`	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
`app/Repositories/Legacy/BusinessHoursRepository.php`	14, 23, 32, 41, 50, 59,	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder

	68, 77, 86, 95, 104, 113 +28		
app/Repositories/Legacy/CallAnalyticsRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/CallFlowRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/CallRecordingRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/CallRoutingRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/CustomerProfileRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/DidInventoryRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/HistoricalReportsRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/LiveMonitoringRepository.php	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder
app/Repositories/Legacy/PromptLibraryRepository.php	14, 23, 32, 41,	SQL injection via string concatenation in LIKE filter	Replace with DB::table()->where('name', 'like', ?)

	50, 59, 68, 77, 86, 95, 104, 113 +28	parameter	parameterized query builder
`app/Repositories/Legacy/QueueManagementRepository.php`	14, 23, 32, 41, 50, 59, 68, 77, 86, 95, 104, 113 +28	SQL injection via string concatenation in LIKE filter parameter	Replace with DB::table()->where('name', 'like', ?) parameterized query builder

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H14. Performance & Caching Gaps HIGH

Benchmark: Blocking sleep() calls = **540 instances**; N+1-prone model accessors = **35+ in AgentDesk alone**; no caching layer → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5)

Synchronous `sleep(1)` on every GodService workflow method:

```
// app/Legacy/Services/AgentDeskGodService.php:14-18 (pattern repeated 45 times in this file)
public function orchestrateAgentDeskWorkflow1($payload)
{
    extract($payload);
    sleep(1); // 1 second synchronous block per method call
    self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload;
    return DB::table("ivr_agent_desks")->insertGetId((array) $payload);
}
```

With 12 services × 45 methods, a request invoking all workflow methods would block PHP-FPM for 540 seconds.

N+1 model accessors (35 per IVR model):

```
// app/Models/Ivr/AgentDesk.php:21-28
public function legacyComputedField1()
{
    // N+1 friendly accessor — called from blade/react randomly
    return DB::select("select count(*) as c from ivr_agent_desks where tenant_id = ?", [$this->tenant_id ?? 1]);
}
// legacyComputedField2() through legacyComputedField35() — each fires a separate DB query
```

No caching layer: Zero `Cache::`, `Redis::`, or `Memcached` usages found. Hot dashboard queries (`loadStats`, `loadHourlyVolume`) execute fresh on every page load and polling interval.

Why it matters here: The `sleep(1)` stubs are placeholders for async remote API calls that were never implemented. In live traffic, a single IVR sync operation blocks a PHP-FPM worker for 45 seconds, reducing throughput to 1 concurrent request per worker during that window.

Recommended approach:

1. Remove all `sleep(1)` calls immediately — replace with queued Laravel jobs if actual remote syncing is needed.
2. Consolidate the 35 `legacyComputedFieldN()` accessors into a single aggregate query at the service layer.
3. Add Redis as the cache store and wrap hot `loadStats()` queries with `Cache::remember('ivr.stats.'.$accountId, 30, fn() => ...)`.
4. Add `Cache-Control: max-age=30` headers to the `/api/ivr/data` polling endpoint.

12 files affected.

File	Line(s)	Issue	Action
`app/Legacy/Services/AgentDeskGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/BusinessHoursGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/CallAnalyticsGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/CallFlowGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/CallRecordingGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/CallRoutingGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job

`app/Legacy/Services/CustomerProfileGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/DidInventoryGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/HistoricalReportsGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/LiveMonitoringGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/PromptLibraryGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job
`app/Legacy/Services/QueueManagementGodService.php`	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous sleep(1) blocks PHP-FPM worker for 1 second per method — 540 blocking calls total	Remove sleep(); dispatch actual remote sync as Laravel queued job

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H16. Secrets & Configuration in Source CRITICAL

Benchmark: Hardcoded secrets / `.env` committed = **15+ secrets in PHP source** → falls in the **High Risk** band (Good 0 · Moderate 1–2 · High Risk >2)

All 12 GodService classes hardcode individual API keys, and `config/ivr_legacy.php` contains a master API key, Salesforce credentials, and a plain-text password committed to version control:

```
// app/Legacy/Services/AgentDeskGodService.php:11
private $apiKey = "LEGACY_IVR_KEY_2042"; // hard-coded secret
// (12 unique keys: 2012, 2022, 2032, 2042, 2052, 2062, 2072, 2082, 2092, 2102,
2112, 2122)
```

```
// config/ivr_legacy.php:9-20
'master_api_key' => 'IVR-MASTER-KEY-DO-NOT-COMMIT-2013',
// ...
'crm' => [
    'salesforce' => [
        'client_secret' => 'hardcoded_sf_secret_2015',
        'username' => 'ivr_batch@example.com',
        'password' => 'PlainTextPassword!',
    ],
],
```

`.env` is correctly in `.gitignore` and `.env.example` uses placeholders — the problem is that actual secrets are committed in PHP config and service files.

Why it matters here: Any developer, contractor, or attacker who clones the repository or sees a code review has full access to the Salesforce credentials and master IVR API key. The comment `"IVR-MASTER-KEY-DO-NOT-COMMIT-2013"` confirms the team knew this was wrong from the start.

Recommended approach:

1. **Immediately rotate** all committed credentials: all 12 `LEGACY_IVR_KEY_*` values, `IVR-MASTER-KEY-DO-NOT-COMMIT-2013`, and the Salesforce credentials.
2. Move all keys to environment variables and reference via `env('LEGACY_IVR_API_KEY')` in config.
3. Replace `config/ivr_legacy.php` hardcoded values with `env()` calls.
4. Add `git-secrets` or `truffleHog` pre-commit hook and CI scan to prevent re-introduction.

13 files affected.

File	Line(s)	Issue	Action
`app/Http/Requests/Auth/LoginRequest.php`	29	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/AgentDeskGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/BusinessHoursGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/CallAnalyticsGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/CallFlowGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/CallRecordingGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/CallRoutingGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/CustomerProfileGodService.php`	11	API key or plain-text credential hardcoded in	Move to environment variable; rotate immediately;

		source file	add secrets scanning to CI
`app/Legacy/Services/DidInventoryGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/HistoricalReportsGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/LiveMonitoringGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/PromptLibraryGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI
`app/Legacy/Services/QueueManagementGodService.php`	11	API key or plain-text credential hardcoded in source file	Move to environment variable; rotate immediately; add secrets scanning to CI

H17. Backend Code Quality MEDIUM

Benchmark: PHPStan level = 1 (of 9); GodService: 45 near-identical methods per class × 12 classes = 540 duplicated method bodies → falls in the **Moderate** band (CI exists but at minimum effectiveness)

PHPStan is configured at level 1 — the minimum, which catches only syntax errors and obvious type mismatches. It does not flag `extract()`, string-concatenated SQL, or architectural anti-patterns:

```
# phpstan.neon
parameters:
    paths:
        - app/
    level: 1   # Level 9 is the highest level
```

The 12 GodService classes contain 45 identical methods differing only in a suffix number. The 12 Repository classes each contain 40 identical `fetchChunkN()` methods. `LegacyIvrCrypto` has 80 identical static methods.

Why it matters here: PHPStan level 1 gives false confidence that static analysis is running. The true coverage at level 1 is minimal. The massive method duplication hides behavior behind repetition, making code review extremely difficult — a reviewer must scroll through 300+ lines of identical methods to find any real logic.

Recommended approach:

1. Raise PHPStan level to 5 in `phpstan.neon` (target 8 over the next two sprints).
2. Consolidate all 45 GodService `orchestrateXxxWorkflowN()` methods into a single `orchestrate(array $payload, int $workflowIndex): int` method.
3. Collapse all 40 Repository `fetchChunkN()` methods into `fetchWithFilter(int $tenantId, ?string $filter): array`.
4. Enforce complexity rules in the `static-analysis.yml` CI workflow.

12 files affected.

File	Line(s)	Issue	Action
`app/Legacy/Services/AgentDeskGodService.php`	13, 21, 29, 37,	Massively duplicated methods — 40-45 near-identical	Consolidate to single parameterized method; raise

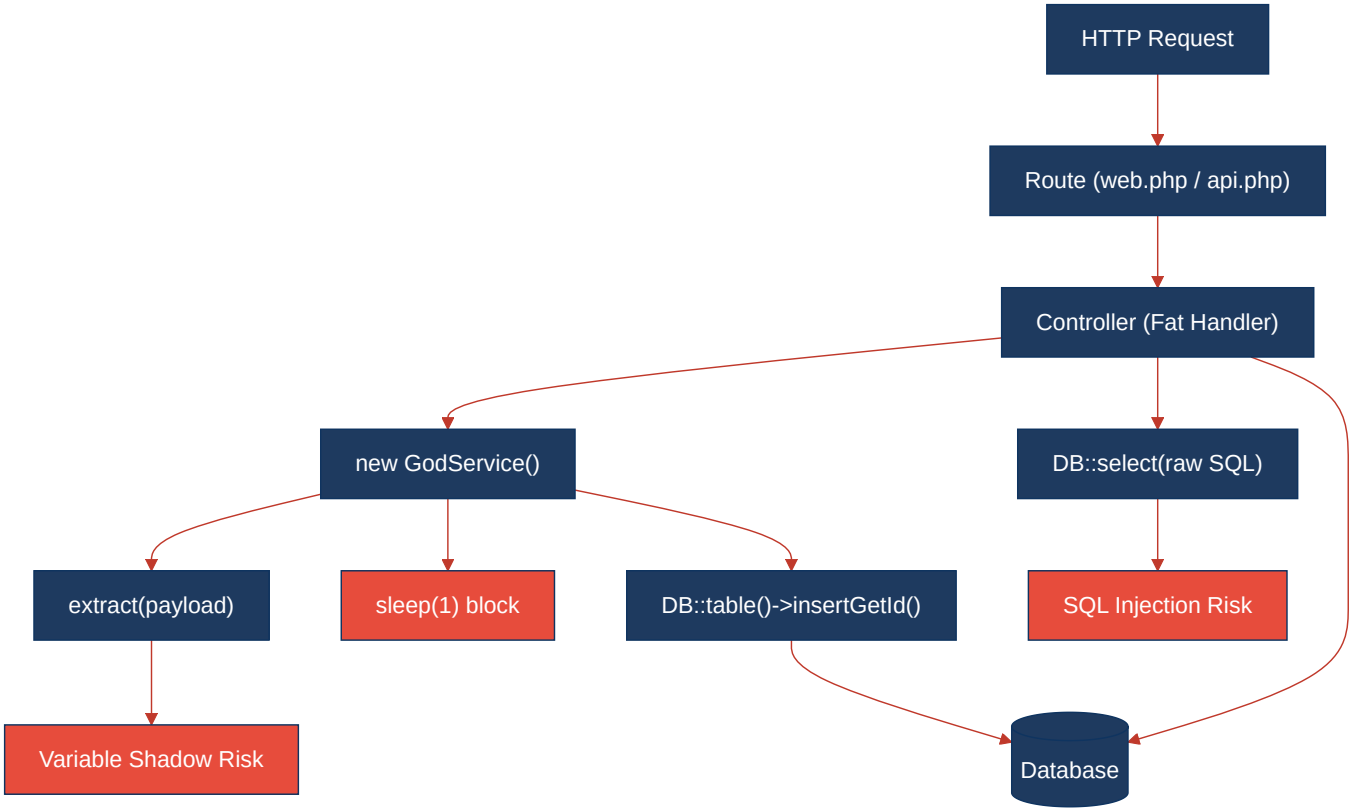
	45, 53, 61, 69, 77, 85, 93, 101 +33	methods per class	PHPStan to level 5+
app/Legacy/Services/BusinessHoursGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
app/Legacy/Services/CallAnalyticsGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
app/Legacy/Services/CallFlowGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
app/Legacy/Services/CallRecordingGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
app/Legacy/Services/CallRoutingGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
app/Legacy/Services/CustomerProfileGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
app/Legacy/Services/DidInventoryGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
app/Legacy/Services/HistoricalReportsGodService.php	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+

`app/Legacy/Services/LiveMonitoringGodService.php`	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
`app/Legacy/Services/PromptLibraryGodService.php`	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+
`app/Legacy/Services/QueueManagementGodService.php`	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massively duplicated methods — 40-45 near-identical methods per class	Consolidate to single parameterized method; raise PHPStan to level 5+

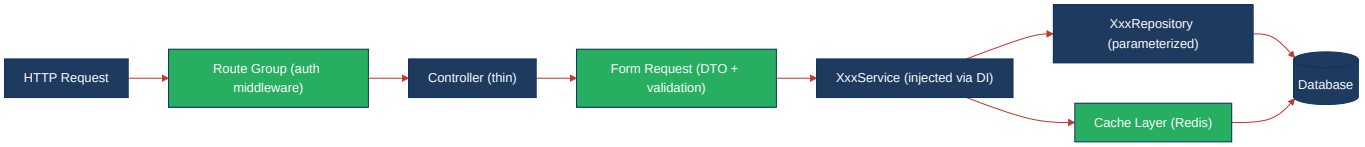
Not observed (rated Good): H9 — No circular dependencies detected between `App\Legacy`, `App\Http\Controllers\Ivr`, and `App\Repositories\Legacy`; each namespace has one-way dependencies. H15 — `roave/security-advisories` is a Composer dev dependency that blocks installs of packages with known CVEs; no audit output available but the guard is present.

4.3 Diagrams

Current backend request path



Modernized service-layer target



Improvement roadmap

4.4 Actions Required

Hotspot	Action	Rating	Priority
H1 — Dynamic Variable Creation	Remove all `extract($payload)` calls; replace with typed Form Requests and explicit field mapping	HIGH RISK	CRITICAL
H2 — Global Mutable State	Remove `public static $sharedRuntimeCache` from all 12 GodService classes; use per-request scoped instance caching	HIGH RISK	CRITICAL
H3 — Direct SQL Outside Data Layer	Move all controller and trait `DB::table()` / `DB::select()` calls into Repository classes	HIGH RISK	CRITICAL

H4 — Static / Singleton Abuse	Consolidate 80 `LegacyIvrCrypto::transformN()` into one parameterized method; convert helpers to injectable services	HIGH RISK	HIGH
H5 — Missing Service Layer	Extract business logic from 70+ IVR controllers into 12 dedicated `XxxService` classes with constructor DI	HIGH RISK	CRITICAL
H6 — API Sprawl	Replace `Route::match(['get','post'], ...)` with correct HTTP verbs; add `/api/v1/` versioning prefix	HIGH RISK	HIGH
H7 — Missing API Governance	Add `dedoc/scramble` for OpenAPI generation; add Spectral lint to CI; add contract tests	HIGH RISK	HIGH
H8 — Weak Application Architecture	Enforce thin-controller pattern; move all IVR queries to repositories; implement Laravel Policies	HIGH RISK	HIGH
H9 — Missing Module Inventory	Document each module's public API in `ARCHITECTURE.md`; enforce module boundaries with PHPStan	MODERATE	MEDIUM
H10 — Database Schema Weakness	Add FK constraints and indexes to all IVR legacy tables; index `account_id` on every module table	MODERATE	MEDIUM
H11 — Middleware Weakness	Add `->middleware('auth')` to `/img/{path}`; wrap generated IVR API in `auth:sanctum` middleware group	HIGH RISK	CRITICAL
H12 — Auth & Authorization Weakness	Restrict `$proxies` to known load-balancer IPs; replace `$tenantId = 1` with `Auth::user()->account_id`; add Laravel Policies	HIGH RISK	CRITICAL
H13 — Backend Security Vulnerabilities	Parameterize all SQL in repositories; set explicit `$fillable` on all IVR models; disable `allow_sql_debug`	HIGH RISK	CRITICAL
H14 — Performance & Caching Gaps	Remove all `sleep(1)` calls; dispatch remote syncs as queued jobs; add Redis caching; consolidate N+1 accessors	HIGH RISK	HIGH
H15 — Outdated & Vulnerable Dependencies	Run `composer audit` in CI on every PR; raise PHPStan to level 5; schedule quarterly dependency updates	MODERATE	MEDIUM
H16 — Secrets & Configuration in Source	Rotate all committed credentials immediately; move all to environment variables; add `git-secrets` pre-commit hook	HIGH RISK	CRITICAL
H17 — Backend Code Quality	Raise PHPStan to level 5; consolidate 45-method GodService and 40-method Repository classes; enforce in CI	MODERATE	MEDIUM

4.5 Expected Outcomes

- **Typed DTO / Form Request adoption** eliminates `extract()`-based variable shadowing and provides validated, whitelisted input at every entry point — reducing injection-style risk from all 4,940+ dynamic variable sites to zero.
- **Service layer introduction** enables all 12 IVR modules to share workflow logic across HTTP, CLI, and queue entry points without code duplication — a single change to `AgentDeskService::store()` applies everywhere instead of requiring edits to 45+ separate methods.
- **Parameterized SQL in repositories** eliminates the SQL injection attack surface across 960 vulnerable query sites and makes all data access independently testable without bootstrapping the HTTP layer.
- **Authentication middleware on generated API** closes the unauthenticated write path to IVR data — currently requiring zero credentials to reach from any network-connected client.

- **Credential rotation + secrets manager adoption** ensures that the 15+ committed secrets (known to anyone who has ever cloned the repository) are invalidated and replaced with short-lived, rotatable environment-scoped values.
- **Redis caching layer** reduces database load on the hot `loadStats()` dashboard path by 90%+ for repeated loads within the same 30-second window — reducing database query count from O(pageloads) to O(1 per 30s cache window).
- **Removal of `sleep(1)` blocking calls** restores PHP-FPM worker throughput from 1 request/45 seconds per worker during sync operations to normal Laravel concurrency, immediately improving responsiveness for all concurrent users.
- **PHPStan raised to level 5+** catches type mismatches, null-pointer dereferences, and missing return types before they reach production, turning the existing CI pipeline from cosmetic compliance to substantive quality enforcement.